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The Gut-First Door

Dietary Neu5Gc, Innate Dietary Agonists and a Proposed Antibody-Defined Subtype of Gut-Initiated Parkinson's Disease

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STATUS OF THIS PAPER

This paper is a hypothesis-generating proposal, not a report of new data. It separates what is **established** in the published literature from what is **hypothesised** by the SGIC framework, and flags points that are **contested**.

It does not propose a cure or a means of eradicating Parkinson's disease. Its claim is narrower: that one inflammatory route into the disease, beginning in the gut, may in some people be driven by antibodies against a dietary sugar, and that this can be tested with existing samples.

Nothing in this paper is medical advice. People living with Parkinson's disease should not change their medication or diet on the basis of it without speaking to their neurologist.

ABSTRACT

A substantial body of evidence now holds that in many people Parkinson's disease begins outside the brain. Alpha-synuclein pathology can appear in the enteric nervous system years before motor symptoms, gut-injected synuclein fibrils reach the mouse brain only when the vagus nerve is intact, and full truncal vagotomy has been associated with lower long-term Parkinson's risk in human registries.[1–4] Gut inflammation appears to matter: people with inflammatory bowel disease develop Parkinson's more often, those treated with anti-TNF drugs appeared markedly protected, and variants in *LRRK2* shift risk for Crohn's disease and Parkinson's disease in the same direction.[7,8]

This paper asks what drives that gut inflammation in the first place. It proposes that, in a subset of people, the Sialic-Glycan Inflammatory Cascade (SGIC) supplies part of the answer. Dietary N-glycolylneuraminic acid (Neu5Gc), which humans cannot make but do incorporate into gut epithelium, becomes a target for circulating anti-Neu5Gc antibodies. Wheat amylase-trypsin inhibitors prime the same mucosa through TLR4, a receptor already implicated in Parkinson's gut pathology.[6,10] The resulting sublytic complement attack and NLRP3 activation would create the chronic local inflammation in which alpha-synuclein misfolds and is carried up the vagus.

The hypothesis is consistent with the repeated association between milk intake and Parkinson's risk, and with the detail of that association once the location of Neu5Gc in milk is taken into account. About three-quarters of cow's milk Neu5Gc is bound to whey proteins and drains away during cheesemaking, which fits the absence of a cheese signal.[16–18,26] Skimming removes fat but not whey protein, so low-fat milk delivers the same protein-bound Neu5Gc as whole milk; the stronger signal for low-fat milk is addressed by a separate, testable hypothesis about the milk fat globule membrane. Six falsifiable predictions are offered, the cheapest of which can be tested on stored serum from existing Parkinson's and inflammatory bowel disease cohorts.

1 Parkinson's Disease Often Begins Outside the Brain

Parkinson's disease is defined clinically by the loss of dopamine-producing neurons in the substantia nigra, but the pathology that precedes that loss is more widespread. In 2003 Braak and colleagues proposed, from autopsy staging, that Lewy pathology appears first in the lower brainstem, particularly the dorsal motor nucleus of the vagus, and the olfactory bulb, and ascends in a predictable sequence. They later added that the enteric nervous system may be involved earlier still.[1,2] Constipation and other gut symptoms commonly precede the diagnosis by many years.

ESTABLISHED **The gut-to-brain route is real in animals and supported in humans**

Mouse model. Kim and colleagues injected preformed alpha-synuclein fibrils into the stomach and duodenal wall of mice. Pathology converted the animals' own synuclein and spread through the vagus into the brainstem and on to the substantia nigra, with neuron loss and motor and cognitive deficits. Truncal vagotomy, or absence of endogenous synuclein, prevented the spread.[4]

Human registry. In a Danish cohort, patients who had undergone full truncal vagotomy had a lower risk of later Parkinson's disease than the general population, most clearly after more than twenty years of follow-up, while superselective vagotomy showed little effect.[3]

Leaky gut in early Parkinson's. Forsyth and colleagues found increased intestinal permeability in newly diagnosed patients, correlating with alpha-synuclein staining in the colonic mucosa and with markers of bacterial endotoxin exposure.[5]

TLR4. Colonic biopsies from Parkinson's patients show increased TLR4-expressing cells, and TLR4-knockout mice are protected from the intestinal inflammation, gut dysfunction and nigral neuron loss produced by oral rotenone.[6]

CONTESTED **Not every case starts in the gut**

Braak's own series included cases that did not follow the staging sequence, and critics have argued that the relationship between synuclein staging and clinical disease is not settled. Current thinking increasingly separates a *body-first* pattern from a *brain-first* pattern.[1,2]

This paper therefore addresses only the gut-initiated route. Any claim it makes applies to that subset and not to Parkinson's disease as a whole.

2 Inflammation as the Engine

If misfolded synuclein can begin in the gut, the next question is what makes the gut a place where it misfolds. Three independent lines of evidence point to inflammation.

ESTABLISHED **Gut inflammation and Parkinson's risk move together**

Inflammatory bowel disease. In administrative claims covering more than 170 million people, those with inflammatory bowel disease were 28% more likely to develop Parkinson's disease than matched controls. Among inflammatory bowel disease patients, those treated with anti-TNF drugs had a 78% lower Parkinson's incidence than those not so treated.[7]

Shared genetics. Hui and colleagues found that variants in *LRRK2*, the most common genetic contributor to Parkinson's disease, raise or lower Crohn's disease risk, and that most variants in the region push both diseases in the same direction.[8]

The inflammasome. Fibrillar alpha-synuclein activates the NLRP3 inflammasome in microglia, inflammasome components are raised in Parkinson's brain tissue, and an oral NLRP3 inhibitor protected dopamine neurons and reduced synuclein pathology in several mouse models.[9]

One caution belongs with the anti-TNF finding. Only two Parkinson's cases occurred among the thirteen thousand treated patients, so the size of the protective effect is imprecise even though its direction is striking.[7] Taken together, however, the three lines make it reasonable to treat chronic gut inflammation as a credible upstream driver of gut-initiated Parkinson's disease. What starts that inflammation in a given person remains largely unexplained. That gap is where the SGIC framework may contribute.

3 The SGIC Framework Applied to the Gut–Brain Axis

3.1 Neu5Gc in the human gut

ESTABLISHED **Humans display a dietary sugar they cannot make, and carry antibodies against it**

Humans lost the *CMAH* gene and cannot synthesise Neu5Gc, yet dietary Neu5Gc from red meat and dairy is taken up and incorporated into human glycans, with accumulation in epithelial linings of hollow organs and in endothelium.[11,12]

Most human sera contain anti-Neu5Gc antibodies, mostly IgG, which are polyclonal and vary widely in quantity and in which Neu5Gc-bearing glycan structures they recognise.[12,15]

Their induction in infancy is linked to non-typeable *Haemophilus influenzae* scavenging dietary Neu5Gc and presenting it to the immune system.[13]

3.2 The four matches at the intestinal mucosa

The SGIC four-match model assigns distinct, non-interchangeable roles to four dietary factors. Applied to the gut wall, they map closely onto what is already known about Parkinson's gut pathology:

- **Neu5Gc, the target.** Incorporated into the glycans of gut epithelium and submucosal capillaries, it provides a surface for anti-Neu5Gc antibodies to bind.
- **Wheat amylase-trypsin inhibitors, the alarm signal.** These wheat proteins activate the TLR4–MD2–CD14 complex on intestinal myeloid cells, and TLR4-deficient mice are protected from the inflammation they cause.[10] TLR4 is the same receptor whose loss protects mice in the rotenone Parkinson's model.[6]
- **Gliadins, the door.** By increasing intestinal permeability, they increase the amount of antigen and bacterial product reaching the submucosa. Increased permeability is documented in early Parkinson's disease.[5]
- **Ammonium compounds, the threshold.** Proposed by the framework as potentiators that lower the threshold at which a given antigen load provokes inflammation. This role rests on the author's clinical observation and has no published mechanistic support yet.

HYPOTHESIS **The central claim of this paper**

In people whose antibody repertoire recognises the Neu5Gc configurations present in their diet, anti-Neu5Gc IgG binding to Neu5Gc-bearing gut epithelium and submucosal vessels fixes complement at sublytic levels. Membrane attack complex inserted without killing the cell is known to trigger NLRP3 inflammasome activation and IL-1β release.[14]

Repeated meal after meal, and primed by TLR4 agonists, this would create a low-grade, persistent inflammatory environment around the enteric neurons of the submucosal and myenteric plexuses: the conditions in which alpha-synuclein is thought to misfold before travelling up the vagus.

On this view, the SGIC does not cause Parkinson's disease directly. It is one candidate for the unexplained gut inflammation that the inflammatory bowel disease, *LRRK2* and TLR4 evidence points to.

3.3 The three gates in the gut

The SGIC three-gate model holds that disease requires all three gates to be open, and that closing any one is sufficient to prevent it. Applied here:

- **Gate One, the antibody repertoire.** The person's anti-Neu5Gc antibodies recognise the bovine-type Neu5Gc glycotopes in their diet. This is measurable.
- **Gate Two, barrier permeability.** Antigen arrives alongside an adjuvant signal, from ATIs, dysbiosis or bacterial endotoxin. This is already documented in early Parkinson's disease.[5]
- **Gate Three, local complement regulation.** In the skin model the framework requires locally reduced CD55 and CD59. The gut has a striking precedent: complete inherited loss of CD55 produces CHAPLE disease, an early-onset protein-losing enteropathy with bowel inflammation and complement hyperactivation, reversed by complement inhibition.[25] This shows that intestinal tissue depends heavily on CD55 to restrain complement. Whether partial, local reductions of CD55 or CD59 occur in the gut of people who later develop Parkinson's disease has not been studied.

Transit time as a Gate Two modifier. Gate Two concerns not only how permeable the barrier is but how long antigen and adjuvant sit against it. In the Honolulu Heart Program, men with less frequent bowel movements in the early 1970s were more likely to develop Parkinson's disease, on average about twelve years later, and commentators at the time suggested that slower transit might allow longer absorption of harmful luminal substances.[27] The SGIC framework gives that suggestion a specific content: slower transit would lengthen the contact of protein-bound Neu5Gc, wheat ATIs and bacterial products with the gut lining after every meal. The direction of cause is uncertain, because Parkinson's pathology damages enteric nerves early and constipation may be the first symptom rather than a cause. The two are not exclusive; slowed transit could begin as a symptom and then deepen the exposure that drives further inflammation. This makes a testable prediction: if transit time modifies exposure, the association between anti-Neu5Gc antibody level and Parkinson's risk (prediction P1) should be strongest in people with long-standing constipation.

The multiplicative structure of the model offers a natural explanation for why most people who eat dairy and wheat, and who carry anti-Neu5Gc antibodies, never develop Parkinson's disease: all three gates must be open at once, for years.

4 The Dairy Signal: Where the Neu5Gc Actually Is

4.1 The epidemiology

ESTABLISHED **Milk intake is consistently associated with higher Parkinson's risk**

A 2014 dose–response meta-analysis of prospective cohorts found Parkinson's risk rose by about 17% for each additional 200 g of milk per day, with a stronger association in men.[17]

In the Nurses' Health Study and Health Professionals Follow-up Study, the association was carried mainly by low-fat dairy, especially skim and low-fat milk, and whole milk showed no association.[16]

A 2026 meta-analysis of ten observational studies again found higher Parkinson's risk with greater dairy and milk intake, while yogurt, cheese, butter and ice cream showed no association.[18]

The absolute risk increase is small, of the order of a few extra cases per 100,000 people per year for each additional daily serving.[17]

On its face, a bovine-specific dietary signal for a gut-initiated disease is what the SGIC framework would predict, and the stronger effect in men sits alongside the higher incidence of Parkinson's disease in men generally. Two details of the pattern look awkward at first: cheese shows no clear signal, and low-fat milk carries more risk than whole milk. Both look different once one asks where the Neu5Gc in milk actually sits.

4.2 Most of the Neu5Gc in milk travels with the whey

ESTABLISHED **Milk sialic acids are mainly protein-bound, and most leave in the whey**

Rinaldi and colleagues measured sialic acids by HPLC across cow, goat and buffalo dairy products. In cow's milk they found roughly 98 mg of Neu5Ac and 3 mg of Neu5Gc per kilogram.[26]

During cheesemaking, about 72% of cow's milk Neu5Gc passed into the whey and only about 28% stayed in the cheese. Most milk sialic acid was bound to whey proteins, notably the glycomacropeptide that rennet cleaves from κ-casein. Ricotta, a cheese made from whey, had the highest sialic acid content of the cheeses tested.[26]

Two consequences follow, and they bear directly on the epidemiology.

4.3 Cheese: from objection to support

Cheese was earlier treated as a problem for the hypothesis because it is concentrated milk solids. In fact, most of the Neu5Gc never reaches the cheese; it drains away in the whey. Per serving, a typical

rennet cheese delivers much less Neu5Gc than its weight in milk solids would suggest. The absence of a cheese signal is therefore what the framework predicts, not an embarrassment to it.

4.4 *Skim versus whole milk: the fat is the variable, not the antigen*

Skimming removes the cream, and with it the milk fat globule membrane. It does not remove whey protein or the glycomacropeptide. A glass of skim milk therefore delivers essentially the same protein-bound Neu5Gc as a glass of whole milk. The two differ in what accompanies the antigen, not in the antigen itself.

This also clarifies a point of transport. In earlier SGIC work the author argued that ganglioside-bound Neu5Gc travels with the milk fat globule membrane through the lymphatics, bypassing the liver. That route may matter for systemic conditions such as acne and visceral adiposity. The gut-initiated Parkinson′s model is different: it needs Neu5Gc presented at the gut lining itself, and for that the protein-bound fraction in the lumen is what counts. The two routes are not in conflict; they serve different target tissues.

> **HYPOTHESIS** **Whole milk brings the antigen but partly closes the door**
>
> The milk fat globule membrane is rich in phospholipids, sphingomyelin and gangliosides. The author proposes that these components support the intestinal barrier, acting on Gate Two of the three-gate model.
>
> If so, whole milk delivers the protein-bound Neu5Gc while also partly protecting the barrier it must cross. Skim milk delivers the same antigen without that protection. The net effect would be a stronger association for low-fat milk, as observed.[16]
>
> This proposal has not been tested and is not supported by any cited study. It is offered because it makes predictions that distinguish it from the alternatives (Section 4.6).

4.5 *How the competing explanations fare*

Explanation	What it predicts for skim versus whole milk	Fit
Pesticide residues	Organochlorine residues dissolve in fat, so whole milk should carry more risk.	Opposite of observed
Lower serum urate	Milk protein lowers urate. Skim and whole milk carry similar protein, so little difference expected.	Does not explain
Galactose	Lactose content is similar in skim and whole milk, so little difference expected.	Does not explain
Residual confounding	People who choose skim milk may differ in other health behaviours.	Possible; check adjustment
SGIC with fat-membrane barrier effect	Same antigen in both; whole milk partly protects the barrier. Cheese spared because Neu5Gc leaves in whey.	Consistent, untested

Table 1. Competing explanations for the milk–Parkinson's association and the low-fat pattern.

Confounding deserves particular attention. Smoking and coffee are both associated with lower Parkinson′s risk, and people who choose low-fat milk may smoke less. Whether the published cohorts adjusted fully for these factors should be confirmed against the original papers before the low-fat signal is treated as biological.

4.6 Remaining difficulties and new predictions

CONTESTED **What the whey account does not yet explain**

Yogurt. Unstrained yogurt keeps its whey, yet yogurt shows no association with Parkinson′s disease. Fermenting bacteria can carry sialidases that cleave terminal sialic acids, which might alter the Neu5Gc glycotopes, but this has not been measured.

Red meat. The richest dietary source of Neu5Gc is not consistently linked to Parkinson′s disease. The configuration argument, that bovine dairy glycotopes differ from those in meat, can accommodate this, but it is an accommodation, not a prediction.

The barrier effect. The proposed protective role of the milk fat globule membrane is the author′s hypothesis and needs direct testing.

The whey account makes three new, inexpensive predictions that do not depend on antibody measurement and can be tested in existing dietary cohorts:

- **Whey protein supplements** concentrate the Neu5Gc-carrying fraction and should show a Parkinson′s signal at least as strong as milk.
- **Ricotta and other whey cheeses** should behave like milk, not like rennet cheese.
- **Strained (Greek-style) yogurt**, which loses most of its whey, should carry less risk than unstrained yogurt.

If whey protein supplements show no association, or Greek yogurt behaves no differently from unstrained yogurt, the whey account would be weakened.

5 What About the Brain Itself?

An obvious question is whether Neu5Gc or anti-Neu5Gc antibodies act directly in the brain. The evidence suggests this is unlikely to be the main route, but not impossible.

ESTABLISHED **The vertebrate brain actively excludes Neu5Gc**

Even in animals that make abundant Neu5Gc in other tissues, brain Neu5Gc is held to trace levels, and this suppression has been conserved across hundreds of millions of years of vertebrate evolution. The small amounts present are mostly on blood-vessel endothelium, not neurons.[19,20]

When Naito-Matsui and colleagues forced Neu5Gc expression in mouse brain, the animals showed abnormal locomotor activity, impaired recognition memory and defective myelination, and became lethally sensitive to a Neu5Gc-binding bacterial toxin.[20]

CONTESTED **Can dietary Neu5Gc reach the brain?**

Taguchi and colleagues injected radiolabelled Neu5Gc into the bloodstream of rats and found that a small amount crossed into the brain, accumulated preferentially in the hippocampus, and was cleared more slowly than Neu5Ac.[21]

This was an intravenous bolus in an animal that makes its own Neu5Gc, not chronic dietary exposure in a human. It shows entry is possible; it does not show it happens to a meaningful degree in people.

The brain endothelium is the one place a direct central effect is plausible, because that is where trace Neu5Gc sits even in Neu5Gc-producing species. Antibody-mediated inflammation of brain capillaries is the subject of a separate SGIC line of argument for vascular disease, but it is not needed here and is not pursued.

A note on gangliosides. Parkinson's disease has an established, separate connection to sialylated glycolipids: GM1 ganglioside, which carries Neu5Ac, is reduced in Parkinson's tissues, and GM1 treatment slowed symptom progression in a small two-year delayed-start trial.[22,23] This does not involve Neu5Gc, and the paper makes no claim linking the two. It is noted only because it shows that sialic acid biology already sits close to the centre of Parkinson's research.

6 Objections

Objection	Response	Status
Many cases are driven by genes (SNCA, GBA, LRRK2) and toxins (pesticides, solvents), not diet.	Agreed. The hypothesis concerns one route in one subtype. LRRK2 itself supports a gut-inflammation route rather than contradicting it.	Answered
The dairy data favour low-fat milk and spare cheese and yogurt.	Most milk Neu5Gc leaves in the whey, explaining cheese. Skim and whole milk carry the same protein-bound Neu5Gc; the fat-membrane barrier effect is proposed for the low-fat pattern (untested). Yogurt remains open.	Partly answered
No study has measured anti-Neu5Gc antibodies in Parkinson's disease.	Correct; a literature search found none. That is the gap this paper proposes to fill, not a point against it.	Open
Anti-Neu5Gc assays are unreliable between laboratories.	A 2024 systematic review confirmed this.[24] Any test must use a matched-pair sialoglycan microarray with a non-mammalian blocker, applied identically to cases and controls.	Answerable
The anti-TNF protection rests on only two cases.	True. The estimate is imprecise; the direction, with IBD and LRRK2 data, is what the argument relies on.	Acknowledged
Dairy and wheat are near-universal; why do so few people develop Parkinson's?	The three-gate model predicts a minority, requiring antibody match, barrier failure and local complement weakness together.	Consistent

Table 2. Principal objections and the author's responses.

7 Testable Predictions

A hypothesis is only useful if a result could disprove it. Table 3 sets out what the framework predicts, in order of cost.

#	Prediction	Test	Result that would falsify it
P1	Anti-Neu5Gc IgG against bovine-type glycotopes is higher in Parkinson′s patients than in matched controls, most clearly in body-first cases with prior constipation or REM sleep behaviour disorder.	Stored serum from an existing Parkinson′s biobank; matched-pair sialoglycan microarray.	No difference, or a difference not confined to body-first cases.
P2	Among people with inflammatory bowel disease, high anti-Neu5Gc titre predicts later Parkinson′s diagnosis.	Nested case-control within an IBD cohort with banked serum.	Titre does not predict Parkinson′s.
P3	The milk–Parkinson′s association is concentrated in people with high bovine-type anti-Neu5Gc antibodies.	Nested case-control in a dietary cohort with stored blood (e.g. NHS / HPFS).	No interaction between milk intake and antibody level.
P4	Colonic biopsies from early Parkinson′s show IgG and C5b-9 deposits co-localising with Neu5Gc near phosphorylated synuclein in the submucosal plexus.	Immunohistochemistry on archived biopsies.	Deposits absent or no different from controls.
P5	In human-like Cmah−/− mice with anti-Neu5Gc antibodies, a Neu5Gc-rich diet accelerates gut-to-brain synuclein spread after gut fibril injection.	Kim et al. model on a Cmah−/− background.	No difference between Neu5Gc-rich and Neu5Gc-free diets.
P6	Whey protein supplements and whey cheeses carry a Parkinson′s signal; strained yogurt carries less than unstrained.	Existing dietary cohorts that record supplement and product type.	No signal for whey products, or no difference between strained and unstrained yogurt.

Table 3. Predictions arising from the hypothesis. P1–P3 and P6 require no new participants.

Prediction P1 is the natural first step. Large Parkinson′s biobanks, including the Parkinson′s Progression Markers Initiative, hold serum from well-characterised patients and controls. A pilot comparison of a few hundred samples, run blind on the matched-pair array used by the Varki and Padler-Karavani laboratories, would cost little and would either open or close this line of inquiry.

8 What This Paper Does Not Claim

- It does not claim that dietary Neu5Gc causes Parkinson′s disease in general, or that most cases arise this way.

- It does not claim that removing dairy, red meat or wheat will prevent, slow or reverse Parkinson's disease. No such evidence exists. Dopamine neurons already lost are not recovered by any known intervention.
- It does not propose eradication. At most, if confirmed, it would identify a measurable risk marker and a modifiable exposure for a subset of gut-initiated cases.
- It does not suggest that people with Parkinson's disease change their treatment. Diet can affect levodopa absorption, and any change should be discussed with the treating neurologist.

9 Conclusion

The evidence that many cases of Parkinson's disease begin in an inflamed gut is now strong. What inflames the gut in those people is not known. This paper proposes that antibodies against a dietary sugar humans cannot make, acting with innate dietary agonists on a permeable gut barrier, are one candidate. The proposal draws on established findings about the vagal route, inflammatory bowel disease, TLR4 and the inflammasome, and, once the whey-bound location of milk Neu5Gc is taken into account, it fits more of the dairy epidemiology than first appeared, though not yet all of it.

It may well be wrong. Its value is that it can be shown to be wrong cheaply: the decisive first measurement can be made on serum that is already sitting in freezers.

Declaration

The author is an independent researcher with no institutional affiliation, no funding for this work and no financial relationship with any food or pharmaceutical manufacturer. The SGIC framework is the author's own work. Correspondence: hamerglycans.com.

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